

Name: _____

Period: _____

Problem Set 3 — Polynomials

“Roots are where polynomials take a nap.”

Quick Reference

Degree: highest exponent with nonzero coefficient. **Max extrema:** at most $\deg - 1$.

Max inflection points: at most $\deg - 2$.

Roots/Zeros: values of x making $p(x) = 0$.

Quadratic formula: $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ for $ax^2 + bx + c = 0$.

1. (4 points) Determine the degree of each polynomial:

(a) $p(x) = 7x^5 - 3x^2 + 11$

(b) $q(x) = -4x^3 + 9x - 6x^4 + 2$

2. (4 points) For a degree- n polynomial, what is the *maximum* number of extrema? What is the *maximum* number of inflection points?

3. (4 points) A polynomial has exactly 5 turning points (extrema). What is the smallest possible degree? What is the smallest possible degree if it also has at least 3 inflection points?

4. (4 points) A degree-7 polynomial has *at most* how many extrema? At most how many points of inflection?

5. (6 points) Let $r(x) = (x - 2)(x + 1)(x - 1)^2$.

(a) List all real roots with multiplicities.

(b) What is the degree of r ? How many roots are there *counting multiplicity*?

6. (6 points) Construct a *monic* cubic with real zeros at -2 and 3 , where 3 is a double root.

7. (6 points) Factor completely over the reals: $x^3 - 4x^2 - 7x + 10$.

8. (6 points) Solve $x^3 - 2x^2 - 9x + 18 = 0$ by factoring (hint: try small integer roots).

9. (6 points) Solve $2x^2 - 5x - 12 = 0$ (use any method).

10. (6 points) Solve $3x^2 + 6x + 5 = 0$ using the quadratic formula. Are the roots real or not?

11. (6 points) Without solving fully, decide whether $x^2 + bx + 25 = 0$ can have *two distinct real* roots if $b = 8$. Explain briefly.

12. (6 points) Given $p(x) = x^4 - 5x^2 + 4$, factor $p(x)$ completely over the reals.

13. (6 points) The polynomial $h(x) = x^3 + ax^2 + bx - 6$ has a root at $x = 2$. Perform the division or use the Factor Theorem to find a factor and then factor $h(x)$ completely (over reals if possible).

14. (6 points) A quartic f has degree 4, exactly two distinct real roots (each simple), and two complex (non-real) roots. What's the maximum possible number of extrema for f ? The maximum possible number of inflection points?

15. (6 points) Suppose $g(x)$ is a cubic with leading coefficient positive and three distinct real roots. Sketch-quality reasoning: as $x \rightarrow -\infty$ and $x \rightarrow +\infty$, what happens to $g(x)$? How many extrema can it have at most?

16. (6 points) Find all real zeros of $x^4 - 13x^2 + 36$.

17. (6 points) Solve $x^3 - 7x^2 + 10x = 0$ and state the multiplicity of each root.

18. (6 points) If k is chosen so that $x = 2$ is a *double root* of $x^3 - 5x^2 + kx - 4 = 0$, find k .

19. (6 points) Let $P(x)$ be a polynomial of degree $n \geq 2$. Explain why it cannot have more than $n - 1$ turning points (a sentence or two is fine).

20. (6 points) True/False (justify briefly):

(a) A degree-2 polynomial can have 2 inflection points.

(b) A degree-5 polynomial can have at most 4 extrema.

21. (6 points) Create a quadratic whose roots are $1 \pm \sqrt{6}$. (Give it in expanded standard form.)

22. (6 points) BONUS (lightly tricky): If $x = -1$ is a triple root of a cubic $T(x)$ and $T(0) = 8$, find a possible formula for $T(x)$ (leading coefficient an integer).